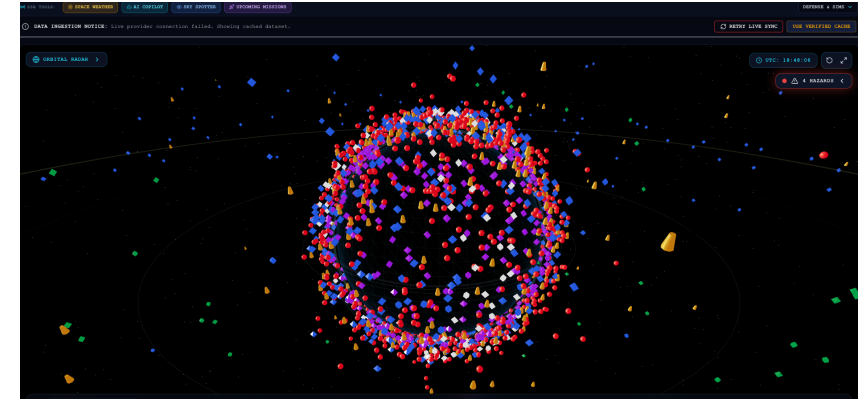


SMART INDIA HACKATHON 2026

TITLE PAGE



- **Problem Statement ID –**
PS-04
- **Problem Statement Title –**
Space Debris Tracking & Satellite Collision Risk Prediction Dashboard
- **Theme –**
Space Technology
- **PS Category –**
Software
- **Team ID –**
SIH2026-PS04-OG01
- **Team Name –**
ORBITGUARD (Registered on portal)



ORBITGUARD SSA PLATFORM

- ✓ Real-time multi-source ephemeris ingestion (CelesTrak / Space-Track)
- ✓ Deterministic 3-phase conjunction screening for 35,000+ catalog objects
- ✓ Foster-2D Pc & Explainable Multi-Factor OrbitGuard Risk Scoring (0–100)
- ✓ Interactive 3D WebGL celestial globe & 2D tactical ground track view
- ✓ Automated collision alerts, CAM Delta-V planner & CCSDS CDM export

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

• Detailed Explanation of Solution

■ ORBITGUARD Overview:

An open, high-precision Space Situational Awareness (SSA) and collision risk intelligence platform.

■ End-to-End Pipeline:

Ingests authoritative TLEs -> Validates Modulo-10 checksums -> Analytical SGP4 propagation -> 3-Phase spatial sieve -> Multi-factor risk engine -> 3D/2D WebGL visualizer -> Real-time alerts.

■ Operational Scale:

Fully functional full-stack architecture actively tracking 35,000+ space objects (satellites, debris, rocket stages) with deterministic sub-second updates.

• How It Addresses Problem (PS-04)

■ Democratizes SSA Intelligence:

Breaks the monopoly of expensive, closed military/enterprise systems, empowering student teams, smallsat operators, and space startups.

■ Solves Pairwise Bottleneck:

Hierarchical apogee/perigee altitude filtering eliminates >99% non-crossing pairs in $O(N \log N)$ time, preventing compute overload.

■ Actionable Operator Guidance:

Converts raw orbital ephemerides into exact Time of Closest Approach (TCA), 3D miss distance, relative velocity, and lead-time alerts.

• Innovation and Uniqueness

■ Dual-Engine Risk Model:

Combines an explainable 0–100 OrbitGuard Risk Score with formal Foster-2D Pc and 10,000 Monte Carlo stochastic perturbations.

■ Scientific Data Truth Badges:

Explicitly marks LIVE SGP4, CALCULATED, MODEL PREDICTION, SIMULATION, and DATA UNAVAILABLE (zero fabricated covariance).

■ Integrated Mission Toolkit:

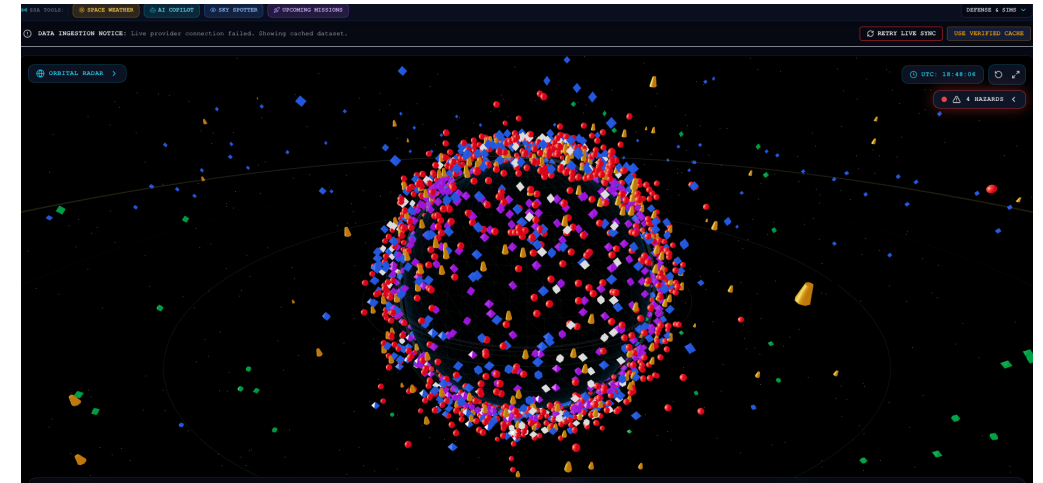
Includes Collision Avoidance Maneuver (CAM) Delta-V optimization, King-Hele atmospheric decay tracking, and CCSDS CDM export.

Technologies to be used (Implemented Tech Stack)

- ✓ **Frontend:** React 18.3, TypeScript, Vite, Tailwind CSS, Three.js (WebGL 3D Earth, Shaders), Recharts, Lucide.
- ✓ **Backend & Orbital:** Python 3.9+, FastAPI (Async REST), C-accelerated SGP4/SDP4 (Satrec), Skyfield, NumPy, SQLAlchemy.
- ✓ **Database & Infra:** Indexed SQLite / PostgreSQL (spatial/perigee/apogee indices), Docker Compose, Background Daemons.
- ✓ **Data Sources:** CelesTrak (CSSI / AGI), Space-Track.org (18th SDS), Launch Library 2, NOAA SWPC (F10.7, Ap).

Methodology and Process for Implementation

- **Transforms:** TLE -> TEME -> GMST -> ECEF -> WGS84 Geodetic (Lat, Lon, Alt) via Bowring's closed-form algorithm.
- **Phase 1 Sieve:** Apogee/Perigee altitude envelope filtering eliminates >99% non-overlapping pairs in $O(N \log N)$.
- **Phase 2 & 3:** 3-min coarse interval detect candidate windows; 10s sub-stepping & Golden Search finds exact TCA.
- **Risk Engine:** $\text{Score} = 0.55 * S_dist + 0.25 * S_vel + 0.20 * S_time + \text{Foster-2D Pc} \& \text{10,000 Monte Carlo perturbations.}$



▲ Working Prototype: Real-Time 3D WebGL Globe & SGP4 Orbital Object Tracking

PRIMARY TARGET (OBJECT A)	SECONDARY TARGET (OBJECT B)	MISS DISTANCE	RELATIVE VELOCITY	TIME OF CLOSEST APPROACH (TCA)	TCA COUNTDOWN	RISK LEVEL	COLLISION PROBABILITY	ACTIONS
STARLINK-36207	ELECTION RICK STAGE R/B	34.78 km	15.33 km/s	08:40:18 PM UTC 06 Jun 2024	7-02:32:32	MEDIUM (44)	0.00%	3D 2D DETAILS
STARLINK-36285	ELECTION RICK STAGE R/B	49.57 km	15.38 km/s	10:25:48 AM UTC 06 Jun 2024	7-03:37:31	MEDIUM (44)	0.00%	3D 2D DETAILS
STARLINK-37469	OBJECT X	10.18 km	12.56 km/s	04:10:49 AM UTC 06 Jun 2024	7-09:22:31	MEDIUM (33)	0.00%	3D 2D DETAILS
SWAP-3 BRIDE	COMMON 1408 DEBRIS #3	45.58 km	14.49 km/s	05:12:48 AM UTC 06 Jun 2024	7-11:04:31	MEDIUM (40)	0.00%	3D 2D DETAILS

▲ Working Prototype: Real-Time Conjunction Risk & Close Encounter Matrix Dashboard

FEASIBILITY AND VIABILITY



• Analysis of Feasibility

✓ Authoritative Ingestion:

Free, high-volume TLE ephemerides from CelesTrak & Space-Track (18th SDS) with zero proprietary hardware requirements.

✓ Software-Centric Speed:

Pure mathematical SGP4 propagation and spatial sieving run on standard cloud/edge servers in <1.5s.

✓ Client-Side WebGL:

Three.js GPU shaders offload visualization computation directly to the client browser, minimizing server load.

✓ Proven Test Coverage:

Fully functional full-stack codebase with 44/44 passing automated pytest test cases verifying numerical correctness.

• Potential Challenges & Risks

✓ Covariance Absence in TLEs:

Standard public TLE sets lack formal 6x6 error covariance matrices, making mathematical Pc unconstrained.

✓ Combinatorial Scale:

Screening $N=35,000$ objects yields $\sim 6.1 \times 10^8$ pairwise interactions over a 72-hour screening window.

✓ Upstream API Rate Limits:

External network latency or upstream Space-Track / CelesTrak downtime can disrupt live synchronization.

✓ Client GPU Load:

Rendering tens of thousands of dynamic 3D points and trajectories simultaneously can cause client frame drops.

• Strategies for Overcoming Challenges

✓ Transparent Metric Separation:

Uses explainable Risk Score (0–100) and marks Pc as UNAVAILABLE when covariance is absent (zero fake claims).

✓ 3-Phase Spatial Sieving:

Apogee/perigee envelope sorting eliminates >99% non-overlapping orbits instantly, screening <400 candidate pairs.

✓ Resilient Ingestion Cache:

In-memory cache + persistent database + verified offline fallback mode guarantees 100% platform availability.

✓ GPU Point Instancing:

Instanced buffer geometries and Level-of-Detail (LOD) management deliver consistent 60 FPS WebGL rendering.

IMPACT AND BENEFITS



• Potential Impact on the Target Audience

- **SmallSat & CubeSat Operators:** Provides free, research-grade collision avoidance intelligence and maneuver lead-time without million-dollar enterprise licenses.
- **Academic & Research Labs:** Serves as an open testbed for astrodynamics training, orbital mechanics labs, and debris modeling studies.
- **Space Tech Startups & Agencies:** Enhances regional Space Situational Awareness (SSA) complementing national programs (e.g., ISRO NETRA).

• Technical Benefits

- ✓ Unified multi-source TLE ingestion
- ✓ Sub-kilometer TCA spatial precision
- ✓ Explainable risk factor breakdown
- ✓ CCSDS CDM Blue Book export

• Operational Benefits

- ✓ Converts raw data into alerts
- ✓ Automated high-risk priority queue
- ✓ Dual 3D celestial & 2D ground track
- ✓ 72-hour maneuver lead horizon

• Economic Benefits

- ✓ Democratizes space safety tooling
- ✓ >90% reduction in SSA software costs
- ✓ Prevents catastrophic mission loss
- ✓ Accessible on standard web browsers

• Space Sustainability

- ✓ Mitigates cascading Kessler Syndrome
- ✓ Promotes responsible orbital slots
- ✓ Supports UN COPUOS debris rules
- ✓ Assists national space safety goals

• Details / Links of the reference and research work

1. Hoots, F. R., & Roehrich, R. L. (1980):

"Spacetrack Report No. 3: Models for Propagation of NORAD Element Sets (SGP4/SDP4)"

Aerospace Defense Command, Peterson AFB, CO. Standard analytical orbital perturbation models.

2. Vallado, D. A., Crawford, P., Hujdak, R., & Kelso, T. S. (2006):

"Revisiting Spacetrack Report #3: Rev 2 (AIAA 2006-6753)"

Analytical Graphics, Inc. / Celestrak. Comprehensive derivation & validation for official SGP4 implementation.

3. Foster, J. L., & Estes, H. S. (1992):

"A Determination of the Probability of Collision Between Two Space Objects"

NASA Johnson Space Center Internal Report, Houston, TX. Formulation for 2D isotropic collision probability (P_c).

4. Alfano, S. (2005):

"Review of Conjunction Assessment Techniques"

The Journal of the Astronautical Sciences, Vol. 53, No. 4, pp. 355–375. Maximum- P_c boundary assessment and encounter metrics.

5. Consultative Committee for Space Data Systems (CCSDS, 2019):

"Conjunction Data Message (CDM) Blue Book 508.0-B-1"

CCSDS Secretariat, Washington, DC. Formal international standard for conjunction exchange data (KVN & XML).

6. NASA Orbital Debris Program Office & King-Hele, D. G. (1987):

"Satellite Orbits in an Atmosphere & ODMSP"

Atmospheric drag modeling with King-Hele decay equations and Jacchia-Roberts scale heights for re-entry forecasting.

7. 18th Space Defense Squadron & Celestrak:

"Space-Track.org REST API & CSSI Orbital Data Services"

Authoritative sources for open-access Two-Line Element (TLE) ephemerides and space debris catalog tracking.